

MATH 231 - Linear Algebra

Homework 1 - Vectors and Complex Numbers

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Part A. Vectors - computation

A.1

(i)-(ii) Use head minus tail and then the Euclidean norm.

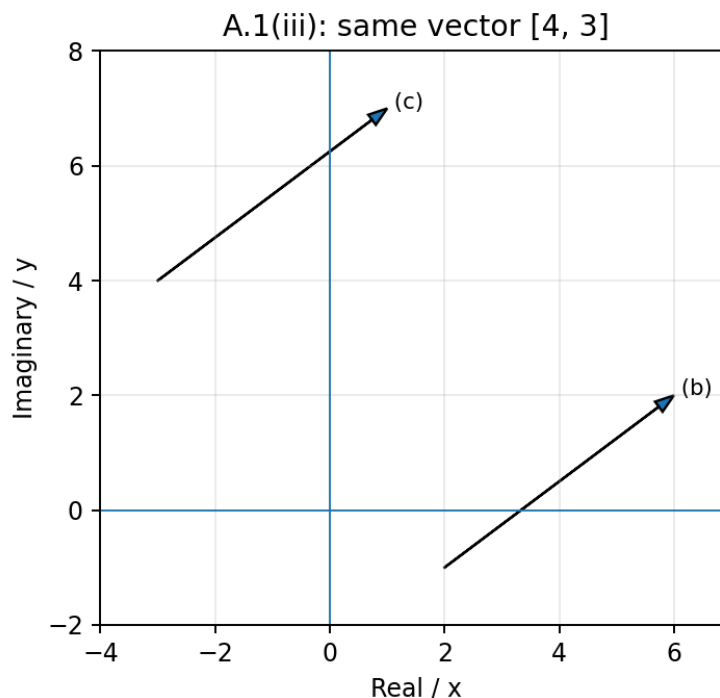
(a) $[5-0, 12-0] = [5, 12]$. $||[5, 12]|| = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

(b) $[6-2, 2-(-1)] = [4, 3]$. $||[4, 3]|| = \sqrt{4^2 + 3^2} = 5$.

(c) $[1-(-3), 7-4] = [4, 3]$. $||[4, 3]|| = 5$.

(d) $[-2-1, 5-1] = [-3, 4]$. $||[-3, 4]|| = \sqrt{(-3)^2 + 4^2} = 5$.

(iii) Rows (b) and (c) are the same vector because both have head-minus-tail equal to $[4, 3]$. Their positions are different, but their magnitude and direction are the same.



(iv) Rows (b), (c), and (d) all have norm 5, but equal norm alone does not make vectors equal. By the definition of vector equality, corresponding components must agree. Here $[4, 3] \neq [-3, 4]$.

A.2

Let $u=[3,-4]$, $v=[-1,2]$, $w=[5,0]$, $\alpha=-3$, $\beta=1/2$.

(a) $u+v = [3+(-1), -4+2] = [2,-2]$.

(b) $u-v = [3-(-1), -4-2] = [4,-6]$.

(c) $v-u = [-1-3, 2-(-4)] = [-4,6]$.

(d) $\alpha u = -3[3,-4] = [-9,12]$.

(e) $\beta v+u = (1/2)[-1,2]+[3,-4] = [-1/2,1]+[3,-4] = [5/2,-3]$.

(f) $2u-3v+w = [6,-8]+[3,-6]+[5,0] = [14,-14]$.

(g) $\|u\|=\sqrt{9+16}=5$; $\|v\|=\sqrt{1+4}=\sqrt{5}$; $\|w\|=\sqrt{25}=5$.

(h) $\|u+v\| = \|[2,-2]\| = \sqrt{8} = 2\sqrt{2}$.

(i) $\|\alpha u\|=\|[-9,12]\|=\sqrt{225}=15$, while $|\alpha|\|u\|=3\cdot 5=15$. Yes, they agree.

A.3

(a) $u+v=[3,4]$, so $\|u+v\|=5$. $\|u\|+\|v\|=3+4=7$. Therefore $5<7$ (strict).

(b) $u+v=[6,9]$, so $\|u+v\|=\sqrt{117}=3\sqrt{13}$. $\|u\|+\|v\|=\sqrt{13}+\sqrt{52}=3\sqrt{13}$. Equality.

(c) $u+v=[-1,2]$, so $\|u+v\|=\sqrt{5}$. $\|u\|+\|v\|=\sqrt{5}+\sqrt{20}=3\sqrt{5}$. Therefore $\sqrt{5}<3\sqrt{5}$ (strict).

(d) $u+v=[2,1]$, so $\|u+v\|=\sqrt{5}$. $\|u\|+\|v\|=\sqrt{5}+0=\sqrt{5}$. Equality.

A.4

(a) Same direction and four times as long: $4v = [8,-12]$.

(b) Opposite direction and half as long: $-(1/2)v = [-1,3/2]$.

(c) Mirror through the origin: $-v=[-2,3]$. Its norm is $\sqrt{13}$. Without computing, $\|-v\|=|-1|\|v\|=\|v\|$, so reflection does not change length.

(d) $[3,-4.5]=(3/2)[2,-3]$, so it is a scalar multiple. For $[4,-6.1]$, the first component requires scalar 2, but $2(-3)=-6\neq-6.1$, so it is not.

(e) $0v=[0,0]$. The zero vector has magnitude 0 and no direction.

A.5

Every subtraction is first interpreted as addition of an additive inverse.

(a)

$$\begin{aligned} & 3(u+2v)-2(u-v) \\ &= 3u+3(2v) + (-2)(u+(-v)) \text{ (rewrite subtraction as addition of an additive inverse; S1)} \\ &= 3u+6v + [(-2)u+(-2)(-v)] \text{ (S3, S1)} \\ &= 3u+6v + (-2)u+2v \text{ (S3; additive-inverse rule)} \\ &= (3+(-2))u + (6+2)v \text{ (A1, A2, S2)} \\ &= u+8v \text{ (S4).} \end{aligned}$$

Check with $u=[1,-2]$, $v=[3,1]$: original = $[25,6]$, and $u+8v=[25,6]$.

(b)

$$\begin{aligned} & -(u-v)+4(v+(1/2)u) \\ &= (-1)(u+(-v)) + 4(v+(1/2)u) \text{ (rewrite subtraction as addition of an additive inverse)} \\ &= -u+v + 4v+2u \text{ (S1, S3)} \\ &= (-1+2)u + (1+4)v \text{ (A1, A2, S2)} \\ &= u+5v \text{ (S4).} \end{aligned}$$

Check: original = $[16,3]$, and $u+5v=[16,3]$.

(c)

$$\begin{aligned} & 5(u+v)-5u-5v \\ &= (5u+5v)+(-5u)+(-5v) \text{ (S1; rewrite subtraction as addition of additive inverses)} \\ &= [5u+(-5u)] + [5v+(-5v)] \text{ (A1, A2)} \\ &= 0+0 = 0 \text{ (A4, A3).} \end{aligned}$$

Check: original = $[0,0]$, and the simplified form $0u+0v=[0,0]$.

A.6

(a) $||\cdot|| : \mathbb{R}^2 \rightarrow \mathbb{R}$. Unary.

(b) $\cdot + \cdot : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Binary.

(c) $\cdot(\cdot) : \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Binary (one scalar and one vector).

(d) $-(\cdot) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Unary.

(e) $\|\cdot - \cdot\| : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$. Binary.

Using the definition of signature from the course notes (domain and codomain), none of these five has the same full signature as another. For example, (b) and (e) both take two vectors as inputs, but (b) returns a vector while (e) returns a real number. So their signatures are different. A signature records the input and output types, but not the rule used to compute the output. The homework statement that two of these five have the same signature appears inconsistent with these five listed operations.

A.7

Scalar: mass, speed, temperature, work, electric charge, distance travelled.

Vector: velocity (direction of motion), force (direction of push/pull), temperature gradient (direction of fastest temperature increase), momentum (direction of motion), displacement (direction from starting point to ending point), acceleration (direction of change of velocity).

(a) The speed is constant at 40 mph. The velocity is not constant because its direction changes around the circle. The car is accelerating because changing velocity direction means the velocity is changing.

(b) Total displacement is the zero vector because the final position equals the starting position. Distance travelled is the positive length of one full lap. They differ because displacement depends only on start and end positions, while distance travelled measures the entire path.

Part B. Vectors - proof

B.1

(a) Let $u=[u_1, u_2]$ and $v=[v_1, v_2]$. Then

$$u-v = [u_1-v_1, u_2-v_2] = [-(v_1-u_1), -(v_2-u_2)] = -(v-u).$$

The middle equality uses ordinary real-number subtraction. Therefore $u-v=-(v-u)$.

(b) Let $u, v, w \in \mathbb{R}^2$. Rewriting subtraction as addition of negatives and using associativity,

$$(u-v)-w = u-v-w.$$

$$[u-(v-w)]-2w = u-v+w-2w = u-v-w.$$

So $(u-v)-w = [u-(v-w)]-2w$ for all u, v, w . For a counterexample, take $u=[0,0]$, $v=[0,0]$, $w=[1,0]$. Then $(u-v)-w=[-1,0]$, while $u-(v-w)=[1,0]$, so subtraction is not associative.

The two bracketings agree exactly when $2w=0$, which over $\mathbb{R}^2$ means $w=0$. If $w=0$, both expressions equal $u-v$. Conversely, if they agree, the displayed identity gives $x=x-2w$, hence $2w=0$ and therefore $w=0$.

B.2

(a) S2. Let $u=[u_1, u_2]$.

$$\begin{aligned}(\alpha+\beta)u &= [(\alpha+\beta)u_1, (\alpha+\beta)u_2] \\&= [\alpha u_1 + \beta u_1, \alpha u_2 + \beta u_2] \quad (\text{distributive law in } \mathbb{R}) \\&= [\alpha u_1, \alpha u_2] + [\beta u_1, \beta u_2] = \alpha u + \beta u.\end{aligned}$$

Thus S2 follows from the distributive law of real numbers.

(b) S3.

$$\begin{aligned}\alpha(\beta u) &= \alpha[\beta u_1, \beta u_2] = [\alpha(\beta u_1), \alpha(\beta u_2)] \\&= [(\alpha\beta)u_1, (\alpha\beta)u_2] = (\alpha\beta)u.\end{aligned}$$

This uses associativity of multiplication in $\mathbb{R}$. S1 and S2 are different: S1 distributes one scalar over a vector sum, while S2 distributes one vector over a sum of scalars.

B.3

Let $v=[v_1, v_2]$ and $\alpha \in \mathbb{R}$. Then

$$\begin{aligned}\|\alpha v\| &= \sqrt{(\alpha v_1)^2 + (\alpha v_2)^2} \\&= \sqrt{(\alpha^2(v_1^2 + v_2^2))} = \sqrt{(\alpha^2)} \sqrt{(v_1^2 + v_2^2)}\end{aligned}$$

$$= |\alpha| \|v\|.$$

We use $\sqrt{(\alpha^2)}=|\alpha|$ because a square root is nonnegative. For $\alpha=-3$ and $v=[1,0]$, the actual norm of $[-3,0]$ is 3. Writing $\sqrt{(\alpha^2)}=\alpha$ would incorrectly give -3, which is impossible because a norm cannot be negative.

B.4

(a) Let $x=0v$. By S2,

$$x=0v=(0+0)v=0v+0v=x+x.$$

$$x+(-x)=(x+x)+(-x).$$

$$0=x+(x+(-x))=x+0=x \quad (\text{A4, A2, A3}).$$

Therefore $0v=0$.

(b) By S4, part (a), and S2,

$$v+(-1)v = 1v+(-1)v = (1+(-1))v = 0v = 0.$$

Therefore $(-1)v$ is an additive inverse of v .

(c) A4 guarantees an additive inverse $-v$ with $v+(-v)=0$. Part (b) shows that $(-1)v$ is also an additive inverse of v . Because the additive inverse is unique, $(-1)v=-v$.

This proof matters because it derives the familiar rule only from the vector-space laws. It therefore remains valid in any vector space, even when vectors are not coordinate pairs.

B.5

Assume $v=\alpha u$ with $\alpha \geq 0$. Then

$$u+v = u+\alpha u = 1u+\alpha u = (1+\alpha)u \quad (\text{S4, S2}).$$

$$\|u+v\| = \|(1+\alpha)u\| = |1+\alpha| \|u\| \quad (\text{B.3}).$$

Since $\alpha \geq 0$, $1+\alpha > 0$, so $|1+\alpha|=1+\alpha$. Hence

$$\|u+v\|=(1+\alpha)\|u\|=\|u\|+\alpha\|u\|=\|u\|+\|\alpha u\|=\|u\|+\|v\|.$$

If $\alpha < 0$, the step $|1+\alpha|=1+\alpha$ is not always valid, and $|\alpha|=\alpha$ is false. For example, take $u=[1,0]$ and $\alpha=-1/2$. Then $v=[-1/2,0]$. We get $\|u+v\|=1/2$, while $\|u\|+\|v\|=3/2$, so the two sides are unequal.

Part C. Complex numbers - computation

C.1

$$z_1 = 4 - 3i: \operatorname{Re}(z_1) = 4, \operatorname{Im}(z_1) = -3.$$

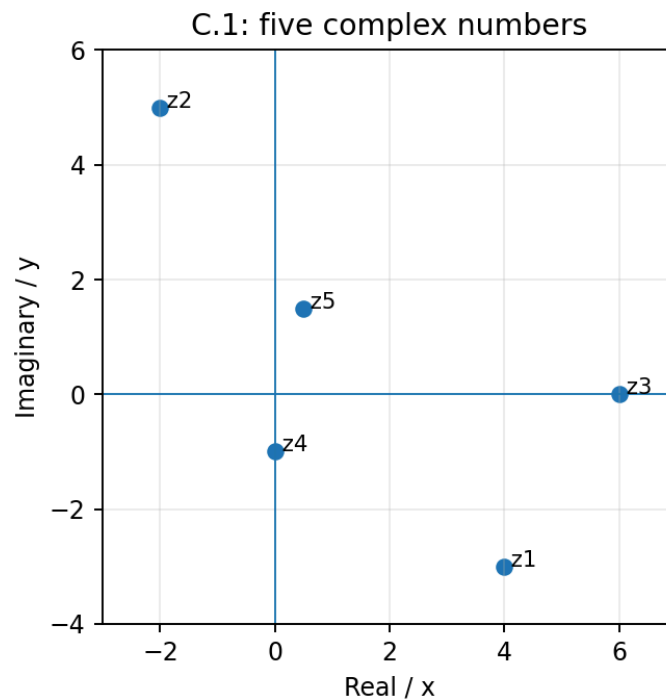
$$z_2 = -2 + 5i: \operatorname{Re}(z_2) = -2, \operatorname{Im}(z_2) = 5.$$

$$z_3 = 6: \operatorname{Re}(z_3) = 6, \operatorname{Im}(z_3) = 0.$$

$$z_4 = -i: \operatorname{Re}(z_4) = 0, \operatorname{Im}(z_4) = -1.$$

$$z_5 = 1/2 + (3/2)i: \operatorname{Re}(z_5) = 1/2, \operatorname{Im}(z_5) = 3/2.$$

z_3 lies on the real axis and is an ordinary real number. z_4 lies on the imaginary axis.



C.2

Let $z = 3 - 2i$ and $w = -1 + 4i$.

(a) $z + w = 2 + 2i$; $z - w = 4 - 6i$; $w - z = -4 + 6i$.

(b) Expanding:

$$zw = (3 - 2i)(-1 + 4i) = -3 + 12i + 2i - 8i^2 = 5 + 14i.$$

Formula check: $ac - bd = 3(-1) - (-2)(4) = 5$, and $ad + bc = 3(4) + (-2)(-1) = 14$, so $zw = 5 + 14i$.

(c) $z^2 = (3 - 2i)^2 = 9 - 12i + 4i^2 = 5 - 12i$.

(d) $\bar{z}=3+2i$, and $z\bar{z}=(3-2i)(3+2i)=13$.

(e) $z/w = ((3-2i)(-1-4i))/((-1+4i)(-1-4i)) = (-11-10i)/17 = -11/17-(10/17)i$.

(f) $1/w = (-1-4i)/17 = -1/17-(4/17)i$.

(g) $|z|=\sqrt{13}$, $|w|=\sqrt{17}$, $|zw|=\sqrt{221}$. Also $|z||w|=\sqrt{13}\sqrt{17}=\sqrt{221}$, so they agree.

C.3

$$i^3=-i, i^4=1, i^5=i, i^{10}=-1, i^{27}=-i, i^{100}=1, (-i)^3=i, (2i)^4=16.$$

The last one is different because $(2i)^4 = 2^4 i^4 = 16(1) = 16$. The extra factor is $2^4 = 16$. Powers of i repeat every four because $i^4=1$; geometrically, multiplying by i makes a 90° counter-clockwise rotation. If $n \bmod 4$ is 0,1,2,3, then i^n is 1,i,-1,-i respectively.

C.4

Let $z=5-12i$.

(a) $\bar{z}=5+12i$. $z\bar{z}=169$. $|z|=13$. Thus $|z|^2=169=z\bar{z}$.

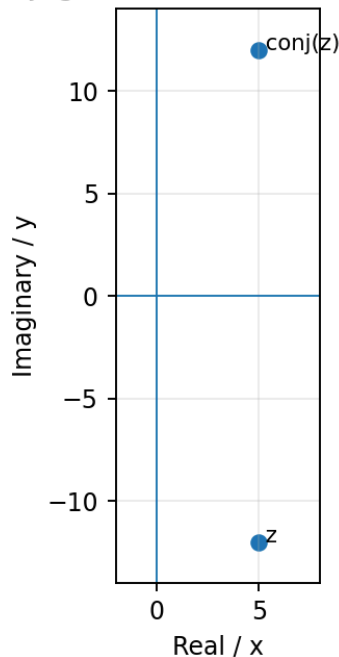
(b) $z+\bar{z}=10=2\text{Re}(z)$.

$z-\bar{z}=-24i=2i \text{Im}(z)$.

General identities: $z+\bar{z}=2\text{Re}(z)$ and $z-\bar{z}=2i \text{Im}(z)$.

(c) z and $\bar{z}$ are reflections of each other across the real axis.

C.4(c): conjugation reflects across the real axis



(d) $\text{conjugate}(\text{conjugate}(z))=z$. Conjugating twice returns to the original point.

(e) $z=\bar{z}$ exactly when $\text{Im}(z)=0$. Therefore the solution set is the real axis, $\mathbb{R} \subset \mathbb{C}$.

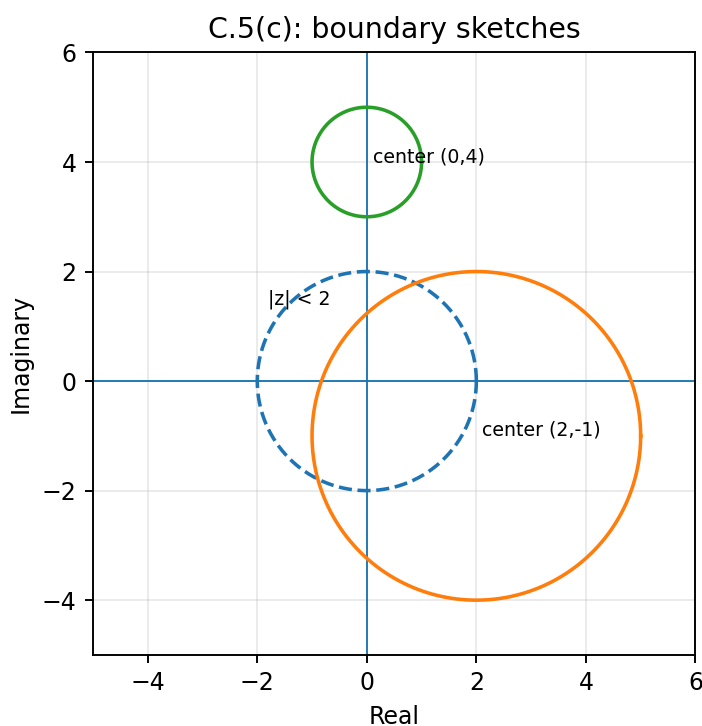
C.5

(a) $|-3+4i|=5$; $|1+i|=\sqrt{2}$; $|-7|=7$; $|2i|=2$; $|0|=0$.

(b) Distance from $3+2i$ to $-1-i$: $|4+3i|=5$. Distance from $5-i$ to $5+3i$: $|-4i|=4$.

For the second pair, both points have the same real coordinate 5, so the distance is just the difference between their imaginary coordinates: $3-(-1)=4$.

(c) $|z|<2$ is the open disc centered at 0 with radius 2; its boundary is excluded. $|z-(2-i)|=3$ is exactly the circle centered at $2-i$ with radius 3, so the boundary circle is included because the set is the circle itself. $|z-4i|\leq 1$ is the closed disc centered at $4i$ with radius 1; its boundary is included.



(d) Closed disc: $|z-(-3+2i)|\leq 5$, equivalently $|z+3-2i|\leq 5$.

(e) $|(1+i)-(2-i)|=|-1+2i|=\sqrt{5}<3$.

So $1+i$ is inside the disc. Its distance to the boundary measured radially is $3-\sqrt{5}$.

(f) $|z|=0$ contains only $z=0$.

C.6

(a) $|i|=1$; $|\sqrt{2}/2+(\sqrt{2}/2)i|=1$; $|1/2+(\sqrt{3}/2)i|=1$; $|-3/5+(4/5)i|=1$.

(b) $i=\cos(\pi/2)+i\sin(\pi/2)$; $\sqrt{2}/2+(\sqrt{2}/2)i=\cos(\pi/4)+i\sin(\pi/4)$;
 $1/2+(\sqrt{3}/2)i=\cos(\pi/3)+i\sin(\pi/3)$.

$-3/5+(4/5)i = \cos(2.2143)+i\sin(2.2143)$, approximately; Quadrant II.

(c) $e^{(i\pi/2)}=i$; $e^{(i\pi)}=-1$; $e^{(2\pi i)}=1$; $e^{(i\pi/4)}=\sqrt{2}/2+(\sqrt{2}/2)i$; $e^{(-i\pi/2)}=-i$.

(d) Euler's identity is $e^{(i\pi)}+1=0$.

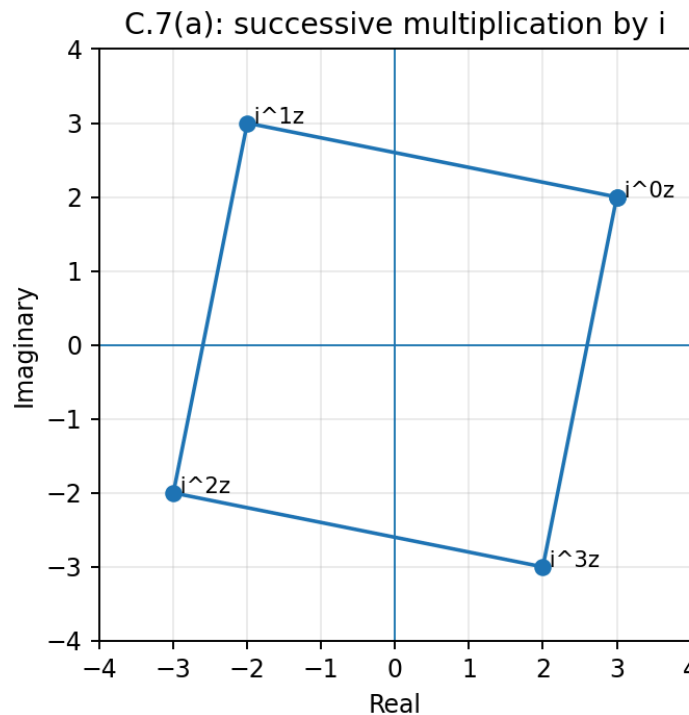
(e) A general complex number needs two real coordinates. On the unit circle the modulus is fixed at 1, so the angle θ alone determines the point.

C.7

Let $z=3+2i$.

(a) $iz=-2+3i$; $i^2z=-3-2i$; $i^3z=2-3i$; $i^4z=3+2i=z$.

The four distinct points form a square centered at the origin.



(b) $|iz|=\sqrt{13}=|z|$. A rotation preserves length.

(c) $(-i)z=2-3i$. The point turns 90° clockwise.

(d) $(4+i)i=-1+4i$, so $(4,1)$ rotates to $(-1,4)$.

(e) Multiply by $-1 = e^{i\pi}$: $(4+i)(-1) = -4-i$, so the point becomes $(-4, -1)$.

(f) $iz = i(a+bi) = -b+ai$. Thus $[a, b]$ becomes $[-b, a]$.

The old second coordinate becomes the negative first coordinate, and the old first coordinate becomes the second: a 90° counter-clockwise rotation.

Part D. Complex numbers - proof

D.1

Write $z = a+bi$ with $a, b \in \mathbb{R}$, so $\bar{z} = a-bi$.

(a) $z + \bar{z} = (a+bi) + (a-bi) = 2a = 2\operatorname{Re}(z)$.

(b) $z - \bar{z} = (a+bi) - (a-bi) = 2bi = 2i \operatorname{Im}(z)$.

(c) If z is real, $b=0$, so $z = \bar{z}$. Conversely, if $z = \bar{z}$, then $a+bi = a-bi$, so $2b=0$ and $b=0$. Therefore z is real iff $z = \bar{z}$.

(d) If $\operatorname{Re}(z) = 0$, then $a=0$, so $z = bi$ and $\bar{z} = -bi = -z$. Conversely, if $\bar{z} = -z$, then $a-bi = -a-bi$, so $2a=0$ and $a=0$. Thus $\operatorname{Re}(z) = 0$ iff $\bar{z} = -z$.

D.2

Write $z = a+bi$ and $w = c+di$.

(a) $\operatorname{conjugate}(z+w) = \operatorname{conjugate}((a+c) + (b+d)i) = (a+c) - (b+d)i = (a-bi) + (c-di) = \bar{z} + \bar{w}$.

(b) $zw = (ac-bd) + (ad+bc)i$, so $\operatorname{conjugate}(zw) = (ac-bd) - (ad+bc)i$.

$$\bar{z}\bar{w} = (a-bi)(c-di) = (ac-bd) - (ad+bc)i.$$

Therefore $\operatorname{conjugate}(zw) = \bar{z}\bar{w}$. Geometrically, reflecting a product across the real axis gives the same result as reflecting both factors first and then multiplying them.

D.3

Let $z = a+bi$ and $w = c+di$. Then $zw = (ac-bd) + (ad+bc)i$.

(a) $|zw|^2 = (ac-bd)^2 + (ad+bc)^2$
 $= a^2c^2 - 2abcd + b^2d^2 + a^2d^2 + 2abcd + b^2c^2.$

(b) The terms $-2abcd$ and $+2abcd$ cancel. Thus

$$|zw|^2 = a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 = (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2.$$

Both sides are nonnegative, so taking the nonnegative square root gives $|zw| = |z||w|$.

D.4

$$(a) |\cos\theta + i \sin\theta| = \sqrt{(\cos^2\theta + \sin^2\theta)} = \sqrt{1} = 1.$$

The identity used is $\cos^2\theta + \sin^2\theta = 1$.

(b) Euler's formula gives $e^{i\theta} = \cos\theta + i \sin\theta$. Therefore

$$|e^{i\theta}| = |\cos\theta + i \sin\theta| = 1.$$

D.5

(a) By multiplicativity of modulus and D.4(b),

$$|z e^{i\theta}| = |z| |e^{i\theta}| = |z| \cdot 1 = |z|.$$

(b) For $\theta = \pi/2$, $e^{i\pi/2} = i$. Hence

$$|iz| = |z e^{i\pi/2}| = |z|.$$

Part E. Dot product, cosine similarity, and projection

E.1

(a) $\langle u, v \rangle = 3(2) + (-1)(5) = 1$. Also $\langle v, u \rangle = 2(3) + 5(-1) = 1$. They agree by D1 (symmetry).

(b) $v + w = [-2, 7]$. $\langle u, v + w \rangle = 3(-2) + (-1)(7) = -13$. Also $\langle u, v \rangle + \langle u, w \rangle = 1 + [3(-4) + (-1)(2)] = 1 - 14 = -13$. They agree by D2 (additivity).

(c) $\alpha u = [-6, 2]$. $\langle \alpha u, v \rangle = (-6)(2) + 2(5) = -2$. $\alpha \langle u, v \rangle = (-2)(1) = -2$. Also $\langle u, \alpha v \rangle = \langle [3, -1], [-4, -10] \rangle = -12 + 10 = -2$. D3 (homogeneity) gives the first equality directly; D1 together with D3 gives the same scaling rule in the second slot.

(d) $\langle u, u \rangle = 3^2 + (-1)^2 = 10$. $\|u\|^2 = (\sqrt{10})^2 = 10$. This is consistent with D4 (positive definiteness).

(e) $u + v = [5, 4]$, so $\langle u + v, u + v \rangle = 25 + 16 = 41$. Also $\|u\|^2 + 2\langle u, v \rangle + \|v\|^2 = 10 + 2(1) + 29 = 41$.

(f) $\langle u, 0 \rangle = 3(0) + (-1)(0) = 0$. In general $\langle x, 0 \rangle = 0$ because every component is multiplied by 0. This does not contradict D4: D4 says $\langle x, x \rangle > 0$ when $x \neq 0$ (and equals 0 exactly for $x = 0$), not that a dot product with a different vector can never be 0.

(g) $\alpha u + \beta v = -2[3, -1] + (1/2)[2, 5] = [-5, 9/2]$. Thus $\langle \alpha u + \beta v, w \rangle = (-5)(-4) + (9/2)(2) = 29$. Also $\alpha \langle u, w \rangle + \beta \langle v, w \rangle = (-2)(-14) + (1/2)(2) = 29$. D2 and D3 together give this linear-combination rule.

(h) $|\langle u, v \rangle| = |1| = 1$. $\|u\| \|v\| = \sqrt{10} \cdot \sqrt{29} = \sqrt{290} \approx 17.029$. The product of the norms is larger by $\sqrt{290} - 1 \approx 16.029$.

E.2

We prove $|\langle u, v \rangle| \leq \|u\| \|v\|$. If $v=0$, then $\langle u, v \rangle=0$ and both sides are 0, so the inequality holds.

Now assume $v \neq 0$. For any real t , D4 gives $0 \leq \|u+tv\|^2$. Using D1-D3,

$$0 \leq \|u\|^2 + 2t\langle u, v \rangle + t^2\|v\|^2.$$

Choose $t = -\langle u, v \rangle / \|v\|^2$. Then

$$0 \leq \|u\|^2 - \langle u, v \rangle^2 / \|v\|^2.$$

Therefore $\langle u, v \rangle^2 \leq \|u\|^2 \|v\|^2$. Both sides are nonnegative, so taking square roots gives $|\langle u, v \rangle| \leq \|u\| \|v\|$. This proves the Cauchy-Schwarz inequality.

E.3

(a) $u=[3,-1]$, $v=[2,6]$. $\langle u, v \rangle=6-6=0$, $\|u\|=\sqrt{10}$, $\|v\|=\sqrt{40}=2\sqrt{10}$. $\cos(u, v)=0$, so $\theta=90^\circ$ and the vectors are orthogonal.

(b) $u=[2,3]$, $v=[6,9]$. $\langle u, v \rangle=12+27=39$, $\|u\|=\sqrt{13}$, $\|v\|=\sqrt{117}=3\sqrt{13}$. $\cos(u, v)=39/(\sqrt{13} \cdot 3\sqrt{13})=1$, so $\theta=0^\circ$ and the vectors are parallel.

(c) $u=[-4,2]$, $v=[6,-3]$. $\langle u, v \rangle=-24-6=-30$, $\|u\|=\sqrt{20}=2\sqrt{5}$, $\|v\|=\sqrt{45}=3\sqrt{5}$. $\cos(u, v)=-30/(2\sqrt{5} \cdot 3\sqrt{5})=-1$, so $\theta=180^\circ$ and the vectors are antiparallel.

(d) $u=[2,2]$, $v=[0,5]$. $\langle u, v \rangle=10$, $\|u\|=\sqrt{8}=2\sqrt{2}$, $\|v\|=5$. $\cos(u, v)=10/(10\sqrt{2})=1/\sqrt{2}=\sqrt{2}/2$, so $\theta=45^\circ$; this pair is none of parallel, antiparallel, or orthogonal.

(i) In (b), $v=3u$, so $\alpha=3>0$ and the vectors are parallel. In (c), $v=(-3/2)u$, so $\alpha=-3/2<0$ and the vectors are antiparallel. The sign tells whether the vectors point in the same or opposite direction.

(ii) In (b), replacing v by $100v$ gives

$\cos(u, 100v)=\langle u, 100v \rangle / (\|u\| \|100v\|)=100\langle u, v \rangle / (100\|u\| \|v\|)=1$. The value does not change because positive scaling changes length but not direction, so cosine similarity measures relative direction/angle rather than magnitude.

E.4

(a) $u=[4,3]$, $v=[1,0]$. $\langle v, u \rangle=4$ and $\|v\|^2=1$. $\text{comp}_v(u)=4$. $\text{proj}_v(u)=[4,0]$. $u-\text{proj}_v(u)=[0,3]$, and $\langle v, [0,3] \rangle=0$.

(b) $u=[3,4]$, $v=[2,1]$. $\langle v, u \rangle=10$ and $\|v\|^2=5$. $\text{comp}_v(u)=10/\sqrt{5}=2\sqrt{5}$. $\text{proj}_v(u)=(10/5)[2,1]=[4,2]$. $u-\text{proj}_v(u)=[-1,2]$, and $\langle v, [-1,2] \rangle=-2+2=0$.

(c) $u=[-2,5]$, $v=[3,-1]$. $\langle v, u \rangle=-11$ and $\|v\|^2=10$. $\text{comp}_v(u)=-11/\sqrt{10}$. $\text{proj}_v(u)=(-11/10)[3,-1]=[-33/10, 11/10]$. $u-\text{proj}_v(u)=[13/10, 39/10]$, and $\langle v, [13/10, 39/10] \rangle=39/10-39/10=0$.

(d) $u=[2,6]$, $v=[3,-1]$. $\langle v, u \rangle=6-6=0$ and $\|v\|^2=10$. $\text{comp}_v(u)=0$. $\text{proj}_v(u)=[0,0]$. $u-\text{proj}_v(u)=[2,6]$, and $\langle v, [2,6] \rangle=6-6=0$.

E.5

By D1-D4,

$$||u+v||^2 = \langle u+v, u+v \rangle = ||u||^2 + 2\langle u, v \rangle + ||v||^2.$$

Since $\langle u, v \rangle \leq |\langle u, v \rangle|$ and E.2 gives $|\langle u, v \rangle| \leq ||u|| ||v||$,

$$||u+v||^2 \leq ||u||^2 + 2||u|| ||v|| + ||v||^2 = (||u|| + ||v||)^2.$$

Norms are nonnegative, so taking square roots preserves the inequality. Therefore $||u+v|| \leq ||u|| + ||v||$, which proves the triangle inequality.

E.6

Let $c = \langle v, u \rangle / ||v||^2$, so $w = cv$. Using D2 and D3,

$$\langle v, u-w \rangle = \langle v, u-cv \rangle = \langle v, u \rangle - c\langle v, v \rangle.$$

By D4, $\langle v, v \rangle = ||v||^2$, so

$$\langle v, u-w \rangle = \langle v, u \rangle - (\langle v, u \rangle / ||v||^2) ||v||^2 = 0.$$

Therefore v is orthogonal to $u-w$. The assumption $v \neq 0$ makes $||v||^2$ nonzero, so c is defined.